

DAY-4 CSV FILE

```
jupyter csvfiles DAY-4 Last Checkpoint: yesterday
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JupyterLab Python [conda env:base] Anaconda Toolbox

[2]: import pandas as pd
      #read csv file
      df=pd.read_csv('people-100.csv')
      print(df)

      Index      User Id First Name Last Name Sex \
0      1  88F7B33d2bcf9f5  Shelby  Terrell  Male
1      2  f90cD3E76f1A9b9  Phillip  Summers  Female
2      3  DbeAb8CdcfeFC2c  Kristine  Travis  Male
3      4  A318ee3c201ef58  Yesenia  Martinez  Male
4      5  1ba7A3dc874da3c  Lori  Todd  Male
..      ..      ..      ..      ..
95     96  5efda7ca4eB260E  Dennis  Barnes  Female
96     97  CcBfc93d3720bE  Steve  Patterson  Female
97     98  2fec528fAF0b69  Wesley  Bray  Male
98     99  Adc7ad9B6e4A1fe  Summer  Oconnell  Female
99    100  b8D0ad3490FC7e1  Mariah  Bernard  Male

      Email      Phone Date of birth \
0  elijah57@example.net  001-084-906-7849x73518  1945-10-26
1  bethany14@example.com  214.112.6044x4913  1910-03-24
2  bthompson@example.com  277.609.7938  1992-07-02
3  kaitlinkaiser@example.com  584.094.6111  2017-08-03
4  buchananmanuel@example.net  689-207-3558x7233  1938-12-01
..      ..      ..
95  bmartin@example.org  001-095-524-2112x257  1954-07-30
96  latasha46@example.net  001-865-478-5157  1932-04-29
97  reginal1@example.org  995-542-3004x76800  1994-12-28
98  alexiscantrell@example.org  001-273-685-6932x092  2012-04-12
99  pcpoland@example.org  (341)594-6554x44657  2016-11-15

      Job Title
```

```
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JupyterLab Python [conda env:base] Anaconda Toolbox

      Job Title
0      Games developer
1      Phytotherapist
2      Homeopath
3      Market researcher
4      Veterinary surgeon
..      ..
95     Software engineer
96     Barrister
97     Police officer
98     Broadcast journalist
99     IT sales professional

[100 rows x 9 columns]

[3]: #read only 1st 5 data
      import pandas as pd
      df=pd.read_csv('people-100.csv')
      print(df.head())

      Index      User Id First Name Last Name Sex \
0      1  88F7B33d2bcf9f5  Shelby  Terrell  Male
1      2  f90cD3E76f1A9b9  Phillip  Summers  Female
2      3  DbeAb8CdcfeFC2c  Kristine  Travis  Male
3      4  A318ee3c201ef58  Yesenia  Martinez  Male
4      5  1ba7A3dc874da3c  Lori  Todd  Male

      Email      Phone Date of birth \
0  elijah57@example.net  001-084-906-7849x73518  1945-10-26
1  bethany14@example.com  214.112.6044x4913  1910-03-24
2  bthompson@example.com  277.609.7938  1992-07-02
3  kaitlinkaiser@example.com  584.094.6111  2017-08-03
4  buchananmanuel@example.net  689-207-3558x7233  1938-12-01

      Job Title
```

```
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JupyterLab Python [conda env:base] Anaconda Toolbox

      Job Title
0      Games developer
1      Phytotherapist
2      Homeopath
3      Market researcher
4      Veterinary surgeon

[4]: #read only last 5 data
      import pandas as pd
      df=pd.read_csv('people-100.csv')
      print(df.tail())

      Index      User Id First Name Last Name Sex \
95     96  5efda7ca4eB260E  Dennis  Barnes  Female
96     97  CcBfc93d3720bE  Steve  Patterson  Female
97     98  2fec528fAF0b69  Wesley  Bray  Male
98     99  Adc7ad9B6e4A1fe  Summer  Oconnell  Female
99    100  b8D0ad3490FC7e1  Mariah  Bernard  Male

      Email      Phone Date of birth \
95  bmartin@example.org  001-095-524-2112x257  1954-07-30
96  latasha46@example.net  001-865-478-5157  1932-04-29
97  reginal1@example.org  995-542-3004x76800  1994-12-28
98  alexiscantrell@example.org  001-273-685-6932x092  2012-04-12
99  pcpoland@example.org  (341)594-6554x44657  2016-11-15

      Job Title
95     Software engineer
96     Barrister
97     Police officer
98     Broadcast journalist
99     IT sales professional
```

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Code

[5]:

#reads all the data from starting to end
import pandas as pd
dfspd.read_csv('people-100.csv')
print(df.to_string())

	Index	User Id	First Name	Last Name	Sex	Email	Phone	Date of birth	
Job Title	0	1	88F783d2b2cf9f5	Shelby	Terrell	Male	elijah57@example.net	001-084-906-7849x73518	1945-10-26
Games developer	1	2	f90c03E76f1A9b9	Phillip	Summers	Female	bethany14@example.com	214.112.6044x4913	1910-03-24
Phytotherapist	2	3	0beab8ccdfefC2c	Kristine	Travis	Male	bthompson@example.com	277.689.7938	1992-07-02
Homeopath	3	4	A318ee3c201ef58	Yesenia	Martinez	Male	kaitlinkaiser@example.com	584.094.6111	2017-08-03
Market researcher	4	5	1ba7A3dc874da3c	Lori	Todd	Male	buchananmanuel@example.net	689-207-3558x7233	1938-12-01
Veterinary surgeon	5	6	bF007CDEF508658	Erin	Day	Male	tconner@example.org	001-171-649-9856x5553	2015-10-28
Waste management officer	6	7	bE9EEF34c872AF7	Katherine	Buck	Female	conniecovan@example.com	+1-773-151-6685x49162	1989-01-22
Intelligence analyst	7	8	2EF6A4e77FaEaC	Ricardo	Hinton	Male	wyattbishop@example.com	001-447-699-7998x88612	1924-03-26
Hydrogeologist	8	9	ba0cC4Deef08dE8	Dave	Farrell	Male	nmccann@example.net	603-428-2429x7392	2018-10-06
Lawyer	9	10	8e4fB470FE19bF0	Isaiah	Downs	Male	virginiaterrill@example.org	+1-511-372-1544x8206	1964-09-20
Engineer, site	10	11	Bf08baA03C29b3b	Sheila	Ross	Female	huangcathy@example.com	895.881.4746	2008-03-20
Advertising account executive	11	12	F738c69fB34E62E	Stacy	Newton	Male	rayleroy@example.org	710.673.3213x80335	1980-10-20
Warden/ranger	12	13	C03f0ADddadAdCe	Mandy	Blake	Male	jefferynoble@example.org	(992)466-1305x4947	2007-12-08
stocompatibility and immunogenetics)	13	14	b759b748D1dE80d	Bridget	Nash	Female	mercedes44@example.com	(216)627-8359	2004-06-28

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Code

12	13	C03f0ADddadAdCe	Mandy	Blake	Male	jefferynoble@example.org	(992)466-1305x4947	2007-12-08	Scientist, clinical (hi
13	14	b759b748D1dE80d	Bridget	Nash	Female	mercedes44@example.com	(216)627-8359	2004-06-28	
14	15	1F0B70d5A000AF9	Crystal	Farmer	Male	pmiranda@example.org	+1-024-377-5391	1992-03-09	
15	16	508b061c8308d461	Thomas	Knight	Female	braunpriscilla@example.net	+1-360-880-0766	2006-02-18	
16	17	6DdbA5308fEC48C	Maurice	Rangel	Male	sheenabanks@example.com	(246)187-4969	2004-08-20	
17	18	3110775990f066d	Frank	Meadows	Male	gbrewer@example.org	429.965.3902x4447	2008-09-16	
18	19	7f7f1B4c8c9AFAF	Alvin	Paul	Male	gilbertdonaldson@example.com	219.436.0887x07551	1949-05-12	
19	20	88473e15D5c3cD0	Jared	Mitchell	Female	jcortez@example.com	+1-958-849-6781	1921-01-18	
20	21	b31027f8c200AB	Jacqueline	Norton	Female	carias@example.net	819.309.7679x59173	1952-10-09	
21	22	42f48dA8A1aBaDC	Colleen	Hatfield	Female	fknox@example.org	638.584.1090	1949-10-14	
22	23	c8B8c8FCAC348c	Randy	Barnes	Male	huangbill@example.org	001-960-629-7164x67214	1947-12-30	
23	24	f1f89173353A090	Janice	Rhodes	Female	juarezdominique@example.net	001-249-314-9742x6996	1999-11-01	
24	25	c5809fb33e8bABA	Alfred	Mcneil	Female	cassandramorris@example.com	(468)276-9509x53058	1993-05-28	
25	26	c9f228C2A08EC1E	Sean	Levine	Male	sallymiller@example.net	4915828504	2010-10-09	
26	27	9c1bc7EC53fb7CE	Louis	Payne	Male	bsullivan@example.net	6232695307	1916-01-29	
27	28	4dc50e2A2e3a28	Brittney	Vega	Female	ayalajose@example.net	945-739-8686	1932-10-31	
28	29	66f90636Ebae11	Judy	Buckley	Male	irosales@example.net	001-654-208-1241x52830	1963-07-28	
29	30	F0FE2fAd78f8B5	Norman	Weber	Male	mconrad@example.com	223.002.0429	1957-05-21	

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Code

30	31	5d2f6aFbdCAAG85	Isaiah	Camacho	Female	jinblake@example.org	001-536-544-3367	1966-04-07	
31	32	cDa5F30fCd6dEa	Jacqueline	Gallagher	Male	nsampson@example.net	(247)762-8934	1999-02-25	
32	33	8Ef70Bfca802b68	Bonnie	Andrews	Female	caitlin24@example.net	+1-253-987-2776x9161	1953-12-21	
33	34	6Dwc5B5542f8ed8	Brandon	Schmidt	Female	mconley@example.net	+1-386-673-1465x006	1931-05-12	
34	35	3fB8a7f68e12784	Jackson	Sparks	Female	reynoldsdarryl@example.net	(137)908-3129x65035	1980-11-18	
35	36	035eff508A0f24	Melody	Cook	Male	jeannovak@example.org	(826)792-7381	1963-06-25	
36	37	aa614aE4B7CF0C	Leonard	Hurst	Male	clinton78@example.org	941-038-0427x38800	1938-03-13	
37	38	ACcd95AAe3e6cC	Gene	Rich	Female	luisdeleon@example.org	+1-356-818-6604x89537	1946-08-22	
38	39	b6a35de5CB6fc25	Cynthia	Wiggins	Female	rosariodave@example.org	(110)858-2437x70190	1984-01-27	
39	40	e92A191E345A3A	Tanya	McKinney	Female	vickihouston@example.com	(830)774-9002x086	2003-03-12	
40	41	700Ac8f6Cac3fd	Matthew	Stone	Female	evelyn31@example.org	952-381-6360	2017-08-23	Scientist, clinical (hi
41	42	cFA78BcAf013AE	Kirk	Walsh	Female	stephenfuller@example.org	001-826-496-5529x8661	2009-04-08	
42	43	9ed8C94af7cA22a	Willie	Vang	Female	haleymatthews@example.net	741.168.6854x067	1978-02-02	
43	44	fFe78AA737aDbe2	Miguel	Hill	Female	tyrone56@example.org	5247842945	1930-08-26	
44	45	5f2f3fAa8B0946	Darren	Andrews	Male	lhernandez@example.com	(975)799-4261	1997-10-04	
45	46	6bfccfc3c18C684	Haley	Pugh	Female	molly93@example.org	(746)182-6137x2453	1980-09-16	
46	47	f38D2cF7e7ebdf	Danielle	Estrada	Female	jvang@example.org	(890)374-9518x772	1930-07-09	
47	48	Ee4e8129dC7913A	Becky	Brady	Male	erikmueller@example.org	(390)002-0863	1957-06-27	


```
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[6]: import pandas as pd
dfspd.read_csv('people-100.csv')
print(df['Sex'])

0 Male
1 Female
2 Male
3 Male
4 Male
...
95 Female
96 Female
97 Male
98 Female
99 Male
Name: Sex, Length: 100, dtype: object

[5]: import pandas as pd
dfspd.read_csv('people-100.csv', index_col='User Id')
print(df)

User Id      Index First Name Last Name Sex \
88f7b33d2bcf9f5  1 Shelby Terrell Male
f90cd3e76f1a9b9  2 Phillip Summers Female
DbAb8cCdfefC2c  3 Kristine Travis Male
A31Bee3c201ef58  4 Yesenia Martinez Male
1bA7A3dc874da3c  5 Lori Todd Male
...          ...      ...      ...
SeFda7caAeB260E  96 Dennis Barnes Female
CCbfce93d3720bE  97 Steve Patterson Female
```

```
jupyter csvfiles DAY-4 Last Checkpoint: yesterday

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JupyterLab Python [conda env:base] * Anaconda Toolbox

CcbFce93d3720bE  97 Steve Patterson Female
2Fec528aFAf0b69  98 Wesley Bray Male
Adc7ad9B6e4A1fE  99 Summer Oconnell Female
b8D0aD3490fC7e1  100 Mariah Bernard Male

Email Phone \
User Id
88f7b33d2bcf9f5 elijah57@example.net 001-084-906-7849x73518
f90cd3e76f1a9b9 bethany14@example.com 214.112.6044x4913
DbAb8cCdfefC2c bthompson@example.com 277.609.7938
A31Bee3c201ef58 kaitlinskaiser@example.com 584.094.6111
1bA7A3dc874da3c buchananmanuel@example.net 689-207-3558x7233
...      ...
SeFda7caAeB260E bmartin@example.org 001-095-524-2112x257
CCbfce93d3720bE latasha46@example.net 001-865-478-5157
2Fec528aFAf0b69 reginal1@example.org 995-542-3004x76800
Adc7ad9B6e4A1fE alexiscantrell@example.org 001-273-685-6932x092
b8D0aD3490fC7e1 pccopeland@example.org (341)594-6554x44657

Date of birth Job Title
User Id
88f7b33d2bcf9f5 1945-10-26 Games developer
f90cd3e76f1a9b9 1910-03-24 Phytotherapist
DbAb8cCdfefC2c 1992-07-02 Homeopath
A31Bee3c201ef58 2017-08-03 Market researcher
1bA7A3dc874da3c 1938-12-01 Veterinary surgeon
...      ...
SeFda7caAeB260E 1954-07-30 Software engineer
CCbfce93d3720bE 1932-04-29 Barrister
2Fec528aFAf0b69 1994-12-28 Police officer
Adc7ad9B6e4A1fE 2012-04-12 Broadcast journalist
b8D0aD3490fC7e1 2016-11-15 IT sales professional

[100 rows x 8 columns]
```

```
jupyter csvfiles DAY-4 Last Checkpoint: yesterday

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JupyterLab Python [conda env:base] * Anaconda Toolbox

2Fec528aFAf0b69 1994-12-28 Police officer
Adc7ad9B6e4A1fE 2012-04-12 Broadcast journalist
b8D0aD3490fC7e1 2016-11-15 IT sales professional

[100 rows x 8 columns]

[7]: import pandas as pd
dfspd.read_csv('people-100.csv', index_col='User Id')
print(df.loc['88f7b33d2bcf9f5', ['First Name', 'Job Title']])

First Name      Shelby
Job Title      Games developer
Name: 88f7b33d2bcf9f5, dtype: object

[8]: print(df.loc['SeFda7caAeB260E':'b8D0aD3490fC7e1', ['First Name', 'Job Title']])

User Id      First Name      Job Title
SeFda7caAeB260E  Dennis      Software engineer
CCbfce93d3720bE  Steve       Barrister
2Fec528aFAf0b69  Wesley      Police officer
Adc7ad9B6e4A1fE  Summer      Broadcast journalist
b8D0aD3490fC7e1  Mariah      IT sales professional

[ ]:
```

JSON AND EXCEL FILES

JupyterLab interface showing a Jupyter Notebook with two code cells and their outputs.

Code Cell [1]:

```
import pandas as pd
dfspd.read_json("titanic.json")
df.head(10)
```

Output [1]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	None	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	None	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	None	S
5	6	0	3	Moran, Mr. James	male	NaN	0	0	330877	8.4583	None	Q
6	7	0	1	McCarthy, Mr. Timothy J	male	54.0	0	0	17463	51.8625	E46	S
7	8	0	3	Palsson, Master. Gosta Leonard	male	2.0	3	1	349909	21.0750	None	S
8	9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27.0	0	2	347742	11.1333	None	S
9	10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14.0	1	0	237736	30.0708	None	C

Code Cell [2]:

```
import pandas as pd
dfspd.read_excel("Supermarket-Sales-Sample-Data.xlsx")
df.head(10)
```

Output [2]:

	Unnamed: 0	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6	Unnamed: 7	Unnamed: 8
0	NaN	Excel Sample Data	NaN	NaN	NaN	NaN	NaN	NaN	NaN
1	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2	NaN	Supermarket Sales Data	NaN	NaN	NaN	NaN	NaN	NaN	NaN
3	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
4	NaN	NaN	NaN	NaN	NaN	NaN	NaN	Tax	0.1
5	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
6	NaN	Order No	Order Date	Customer Name	Ship Date	Retail Price (USD)	Order Quantity	Tax (USD)	Total (USD)
7	NaN	1001	2024-01-01 00:00:00	John Smith	2024-01-03 00:00:00	49.99	2	9.998	109.978
8	NaN	1002	2024-01-01 00:00:00	Jane Doe	2024-01-04 00:00:00	29.99	1	2.999	32.989
9	NaN	1003	2024-01-02 00:00:00	Michael Johnson	2024-01-07 00:00:00	99.99	3	29.997	329.967

JupyterLab interface showing a Jupyter Notebook with two code cells and their outputs.

Code Cell [2]:

```
import pandas as pd
dfspd.read_excel("Supermarket-Sales-Sample-Data.xlsx")
df.head(10)
```

Output [2]:

	Unnamed: 0	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6	Unnamed: 7	Unnamed: 8
0	NaN	Excel Sample Data	NaN	NaN	NaN	NaN	NaN	NaN	NaN
1	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2	NaN	Supermarket Sales Data	NaN	NaN	NaN	NaN	NaN	NaN	NaN
3	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
4	NaN	NaN	NaN	NaN	NaN	NaN	NaN	Tax	0.1
5	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
6	NaN	Order No	Order Date	Customer Name	Ship Date	Retail Price (USD)	Order Quantity	Tax (USD)	Total (USD)
7	NaN	1001	2024-01-01 00:00:00	John Smith	2024-01-03 00:00:00	49.99	2	9.998	109.978
8	NaN	1002	2024-01-01 00:00:00	Jane Doe	2024-01-04 00:00:00	29.99	1	2.999	32.989
9	NaN	1003	2024-01-02 00:00:00	Michael Johnson	2024-01-07 00:00:00	99.99	3	29.997	329.967

Code Cell [3]:

```
import pandas as pd
dfspd.read_excel("Supermarket-Sales-Sample-Data.xlsx")
print(df)
```

Output [3]:

```
Unnamed: 0      Unnamed: 1      Unnamed: 2  \
0      NaN      Excel Sample Data      NaN
1      NaN      NaN      NaN
2      NaN      Supermarket Sales Data      NaN
3      NaN      NaN      NaN
4      NaN      NaN      NaN
..      ...      ...      ...
72     NaN      1066      2024-02-02 00:00:00
73     NaN      1067      2024-02-03 00:00:00
74     NaN      1068      2024-02-04 00:00:00
75     NaN      1069      2024-02-04 00:00:00
76     NaN      1070      2024-02-04 00:00:00

Unnamed: 3      Unnamed: 4      Unnamed: 5      Unnamed: 6      Unnamed: 7  \
0      NaN      NaN      NaN      NaN      NaN
1      NaN      NaN      NaN      NaN      NaN
2      NaN      NaN      NaN      NaN      NaN
3      NaN      NaN      NaN      NaN      NaN
4      NaN      NaN      NaN      NaN      NaN
..      ...      ...      ...      ...      ...
72     Sarah Gonzalez      2024-02-06 00:00:00      79.99      2      15.998
73     Matthew Smith      2024-02-06 00:00:00      49.99      3      14.997
74     Emily Johnson      2024-02-05 00:00:00      129.99      1      12.999
75     Daniel Brown      2024-02-08 00:00:00      19.99      4      7.996
76     Jennifer Hernandez      2024-02-07 00:00:00      149.99      1      14.999

Unnamed: 8
0      NaN
1      NaN
2      NaN
3      NaN
```

JupyterLab interface showing a Jupyter Notebook with one code cell and its output.

Code Cell [3]:

```
import pandas as pd
dfspd.read_excel("Supermarket-Sales-Sample-Data.xlsx")
print(df)
```

Output [3]:

```
Unnamed: 0      Unnamed: 1      Unnamed: 2  \
0      NaN      Excel Sample Data      NaN
1      NaN      NaN      NaN
2      NaN      Supermarket Sales Data      NaN
3      NaN      NaN      NaN
4      NaN      NaN      NaN
..      ...      ...      ...
72     NaN      1066      2024-02-02 00:00:00
73     NaN      1067      2024-02-03 00:00:00
74     NaN      1068      2024-02-04 00:00:00
75     NaN      1069      2024-02-04 00:00:00
76     NaN      1070      2024-02-04 00:00:00

Unnamed: 3      Unnamed: 4      Unnamed: 5      Unnamed: 6      Unnamed: 7  \
0      NaN      NaN      NaN      NaN      NaN
1      NaN      NaN      NaN      NaN      NaN
2      NaN      NaN      NaN      NaN      NaN
3      NaN      NaN      NaN      NaN      NaN
4      NaN      NaN      NaN      NaN      NaN
..      ...      ...      ...      ...      ...
72     Sarah Gonzalez      2024-02-06 00:00:00      79.99      2      15.998
73     Matthew Smith      2024-02-06 00:00:00      49.99      3      14.997
74     Emily Johnson      2024-02-05 00:00:00      129.99      1      12.999
75     Daniel Brown      2024-02-08 00:00:00      19.99      4      7.996
76     Jennifer Hernandez      2024-02-07 00:00:00      149.99      1      14.999

Unnamed: 8
0      NaN
1      NaN
2      NaN
3      NaN
```


Index	Name	Date	Time	Person	Date	Time	Score	Count	Score
22	NaN	1016	2024-01-08 00:00:00	Amanda Martinez	2024-01-12 00:00:00		149.99	1	14.999
164,989	23	NaN	1017	2024-01-09 00:00:00	James Johnson	2024-01-14 00:00:00	69.99	2	13.998
153,978	24	NaN	1018	2024-01-09 00:00:00	Laura Brown	2024-01-12 00:00:00	39.99	3	11.997
131,967	25	NaN	1019	2024-01-10 00:00:00	Daniel Smith	2024-01-11 00:00:00	199.99	1	19.999
219,989	26	NaN	1020	2024-01-10 00:00:00	Jennifer Davis	2024-01-14 00:00:00	29.99	5	14.995
164,945	27	NaN	1021	2024-01-11 00:00:00	Michael Garcia	2024-01-14 00:00:00	79.99	2	15.998
175,978	28	NaN	1022	2024-01-11 00:00:00	Amy Hernandez	2024-01-15 00:00:00	49.99	3	14.997
164,967	29	NaN	1023	2024-01-12 00:00:00	Christopher Rodriguez	2024-01-17 00:00:00	129.99	1	12.999
142,989	30	NaN	1024	2024-01-12 00:00:00	Jessica Martinez	2024-01-17 00:00:00	19.99	4	7.996
87,956	31	NaN	1025	2024-01-13 00:00:00	David Wilson	2024-01-17 00:00:00	149.99	1	14.999
164,989	32	NaN	1026	2024-01-13 00:00:00	Sarah Smith	2024-01-14 00:00:00	69.99	2	13.998
153,978	33	NaN	1027	2024-01-14 00:00:00	Matthew Johnson	2024-01-18 00:00:00	39.99	3	11.997
131,967	34	NaN	1028	2024-01-14 00:00:00	Emily Davis	2024-01-19 00:00:00	199.99	1	19.999
219,989	35	NaN	1029	2024-01-15 00:00:00	Daniel Wilson	2024-01-19 00:00:00	29.99	5	14.995
164,945	36	NaN	1030	2024-01-15 00:00:00	Jennifer Martinez	2024-01-19 00:00:00	79.99	2	15.998
175,978	37	NaN	1031	2024-01-16 00:00:00	Michael Smith	2024-01-19 00:00:00	49.99	3	14.997
164,967	38	NaN	1032	2024-01-16 00:00:00	Jessica Johnson	2024-01-17 00:00:00	129.99	1	12.999
142,989	39	NaN	1033	2024-01-17 00:00:00	David Brown	2024-01-19 00:00:00	19.99	4	7.996


```
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[24]: import pandas as pd
dfpd.read_json("titanic.json")
print(df)
df.head()
#Print 1st 5 values

#Check Missing Values
df.isnull()
#Used to identify missing values
print(df.isnull())
#Count missing values for all field
print(df.isnull().sum())
#to find the name of the columns in the data set
print(df.columns)
#Find Rows Having Missing Age
print(df[df["Age"].isnull()])

df.fillna()
#Replace missing values.
#Fill Age with Mean
mean_agesdf['Age'].mean()
df['Age'].fillna(df['Age'].mean())
print(df.isnull().sum())

#Fill Embarked with Most Frequent Values
#By using mode() means the most frequently occurring value in a column
mode_embarkeddf['Embarked'].mode()[0]
df['Embarked'].fillna(mode_embarked)

df.dropna()
#Remove missing data
df.dropna(inplace=True)
```

```
jupyter json and excel_file DAY-4 Last Checkpoint: yesterday
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JupyterLab Python [conda env:base] Anaconda Toolbox

df.dropna()
#Remove missing data
df.dropna(inplace=True)

#Remove Rows
df.dropna()
#Remove column
df.dropna(axis=1)

#Group by
#Average Fare Per Class
print(df.groupby('Pclass')['Fare'].mean())
#Count Survivors by gender
print(df.groupby('Sex')['Survived'].sum())
#Multiple Aggregations
df.agg()
print(df.groupby('Pclass').agg([
    'Age': ['mean', 'min', 'max'],
    'Fare': ['mean', 'sum']
]))

df.transform()
#return same number of rows.
#Add Class average fare
df['Avg_Class_Fare'] = df.groupby('Pclass')['Fare'].transform('mean')
print(df)
print(df.columns)

df.interpolate()
#Estimate missing values.
sampledf.DataFrame([
    'Age': [20, 25, None, 35, 40]
])
sample['Age'].sample[ 'Age' ].interpolate()
```

```
jupyter json and excel_file DAY-4 Last Checkpoint: yesterday
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sample['Age'].sample[ 'Age' ].interpolate()
print(sample)

   PassengerId  Survived  Pclass \
0             1         0       3
1             2         1       1
2             3         1       3
3             4         1       1
4             5         0       3
..          ...         ...     ...
886           887         0       2
887           888         1       1
888           889         0       3
889           890         1       1
890           891         0       3

   Name                               Sex  Age  SibSp \
0   Braund, Mr. Owen Harris             male  22.0    1
1  Cumings, Mrs. John Bradley (Florence Briggs Th... female  38.0    1
2     Heikkinen, Miss. Laina             female  26.0    0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)    female  35.0    1
4     Allen, Mr. William Henry             male  35.0    0
..          ...         ...     ...     ...
886   Montvila, Rev. Juozas             male  27.0    0
887    Graham, Miss. Margaret Edith          female  19.0    0
888  Johnston, Miss. Catherine Helen "Carrie"    female  NaN    1
889    Behr, Mr. Karl Howell             male  26.0    0
890   Dooley, Mr. Patrick                 male  32.0    0

   Parch  Ticket   Fare Cabin Embarked
0      0   A/5 21171   7.2500   None        S
1      0  PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282  7.9250   None        S
3      0  113803  53.1000  C123        S
4      0  373450  8.0500   None        S
```



```
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886 0 211536 13.0000 None S
887 0 112053 30.0000 B42 S
888 2 W./C. 6607 23.4500 None S
889 0 111369 30.0000 C148 C
890 0 370376 7.7500 None Q

[891 rows x 12 columns]
PassengerId Survived Pclass Name Sex Age SibSp Parch Ticket \
0 False False False False False False False False False
1 False False False False False False False False False
2 False False False False False False False False False
3 False False False False False False False False False
4 False False False False False False False False False
.. ..
886 False False False False False False False False False
887 False False False False False False False False False
888 False False True False False False False False False
889 False False False False False False False False False
890 False False False False False False False False False

Fare Cabin Embarked
0 False True False
1 False True False
2 False True False
3 False False False
4 False True False
.. ..
886 False True False
887 False False False
888 False True False
889 False False False
890 False True False

[891 rows x 12 columns]
```

```
jupyter json and excel_file DAY-4 Last Checkpoint: yesterday

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JupyterLab Python [conda env:base] Anaconda Toolbox

[891 rows x 12 columns]
PassengerId 0
Survived 0
Pclass 0
Name 0
Sex 0
Age 177
SibSp 0
Parch 0
Ticket 0
Fare 0
Cabin 687
Embarked 2
dtype: int64
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
       'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='object')
PassengerId Survived Pclass Name \
5 6 0 3 Moran, Mr. James
17 18 1 2 Williams, Mr. Charles Eugene
19 20 1 3 Nassefman, Mrs. Fatima
26 27 0 3 Emir, Mr. Farred Chehab
28 29 1 3 O'Dwyer, Miss. Ellen "Nellie"
.. ..
860 0 3 Razi, Mr. Raheed
863 864 0 3 Sage, Miss. Dorothy Edith "Dolly"
868 869 0 3 van Melkebeke, Mr. Philemon
878 879 0 3 Laleff, Mr. Kristo
888 889 0 3 Johnston, Miss. Catherine Helen "Carrie"

Sex Age SibSp Parch Ticket Fare Cabin Embarked
5 male NaN 0 0 330877 8.4583 None Q
17 male NaN 0 0 244373 13.0000 None S
19 female NaN 0 0 2649 7.2250 None C
26 male NaN 0 0 2631 7.2250 None C
```

```
jupyter json and excel_file DAY-4 Last Checkpoint: yesterday

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19 female NaN 0 0 2649 7.2250 None C
26 male NaN 0 0 2631 7.2250 None C
28 female NaN 0 0 330959 7.8792 None Q
.. ..
859 male NaN 0 0 2629 7.2292 None C
863 female NaN 0 2 CA. 2343 69.5500 None S
868 male NaN 0 0 345777 9.5000 None S
878 male NaN 0 0 349217 7.8958 None S
888 female NaN 1 2 W./C. 6607 23.4500 None S

[177 rows x 12 columns]
PassengerId 0
Survived 0
Pclass 0
Name 0
Sex 0
Age 0
SibSp 0
Parch 0
Ticket 0
Fare 0
Cabin 687
Embarked 2
dtype: int64
Pclass
1 84.154687
2 20.662183
3 13.675550
Name: Fare, dtype: float64
Sex
female 233
male 109
Name: Survived, dtype: int64
Age
mean min max
Fare
mean sum
```

jupyter json and excel_file DAY-4 Last Checkpoint: yesterday

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JupyterLab Python [conda env:base] Anaconda Toolbox

```
Name: Survived, dtype: int64
Age
mean min max Fare
mean sum
Pclass
1 37.048118 0.92 80.0 84.154687 18177.4125
2 29.866958 0.67 70.0 20.662183 3801.8417
3 26.403259 0.42 74.0 13.675550 6714.6951
PassengerId Survived Pclass \
0 1 0 3
1 2 1 1
2 3 1 3
3 4 1 1
4 5 0 3
...
886 887 0 2
887 888 1 1
888 889 0 3
889 890 1 1
890 891 0 3

Name Sex Age \
0 Braund, Mr. Owen Harris male 22.000000
1 Cumings, Mrs. John Bradley (Florence Briggs Th... female 38.000000
2 Heikinen, Miss. Laina female 26.000000
3 Futrelle, Mrs. Jacques Heath (Lily May Peel) female 35.000000
4 Allen, Mr. William Henry male 35.000000
...
886 Montvila, Rev. Juozas male 27.000000
887 Graham, Miss. Margaret Edith female 19.000000
888 Johnston, Miss. Catherine Helen "Carrie" female 29.699118
889 Behr, Mr. Karl Howell male 26.000000
890 Dooley, Mr. Patrick male 32.000000

SibSp Parch Ticket Fare Cabin Embarked Avg_Class_Fare
0 1 0 A/S 21171 7.2500 None S 13.675550
```

jupyter json and excel_file DAY-4 Last Checkpoint: yesterday

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JupyterLab Python [conda env:base] Anaconda Toolbox

```
Doooley, Mr. Patrick male 32.000000

SibSp Parch Ticket Fare Cabin Embarked Avg_Class_Fare
0 1 0 A/S 21171 7.2500 None S 13.675550
1 1 0 PC 17599 71.2833 C85 C 84.154687
2 0 0 STON/O2. 3101282 7.9250 None S 13.675550
3 1 0 113803 53.1000 C123 S 84.154687
4 0 0 373450 8.0500 None S 13.675550
...
886 0 0 211536 13.0000 None S 20.662183
887 0 0 112053 30.0000 B42 S 84.154687
888 1 2 W./C. 6607 23.4500 None S 13.675550
889 0 0 111369 30.0000 C148 C 84.154687
890 0 0 370376 7.7500 None Q 13.675550

[891 rows x 13 columns]
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
       'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked', 'Avg_Class_Fare'],
      dtype='object')

Age
0 20.0
1 25.0
2 30.0
3 35.0
4 40.0

new data set

[44]: #num sent data
import pandas as pd
studentspd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_name="students")
marksspd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_name="marks")
contactspd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_name="contact")
```

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JupyterLab Python [conda env:base] Anaconda Toolbox

```
[44]: #num sent data
import pandas as pd
studentspd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_name="students")
marksspd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_name="marks")
contactspd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_name="contact")

#1. INNER MERGE
#returns only matching Student_IDs from both DataFrames.

inner_df=pd.merge(
    students,
    marks,
    on="Student_ID",
    how="inner"
)

print(inner_df.head())
print(inner_df.shape)

Student_ID Name Department Maths Python
0 1001 Student_1 CSE 61 56
1 1002 Student_2 IT 62 57
2 1003 Student_3 ECE 63 58
3 1004 Student_4 EEE 64 59
4 1005 Student_5 MECH 65 60
(45, 5)

[45]: #2.LEFT MERGE
#keeps all records from the left DataFrame (students).
left_df=pd.merge(
    students,
    marks,
    on="Student_ID",
    how="left"
```

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```

print(left_df.tail())
print(left_df.shape)

Student_ID Name Department Maths Python
45 1046 Student_46 CSE NaN NaN
46 1047 Student_47 IT NaN NaN
47 1048 Student_48 ECE NaN NaN
48 1049 Student_49 EEE NaN NaN
49 1050 Student_50 MECH NaN NaN
(50, 5)

[ ]: #3. RIGHT MERGE
right_dfsp.merge(
    students,
    marks,
    on="Student_ID",
    how="right"
)
print(right_df.tail())
print(right_df.shape)

[43]: #4. OUTER MERGE
outer_dfsp.merge(
    students,
    marks,
    on="Student_ID",
    how="outer"
)
print(outer_df.tail())
print(outer_df.shape)

Student_ID Name Department Maths Python
50 2001 NaN NaN 75.0 80.0

```

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```

52 2003 NaN NaN 75.0 80.0
53 2004 NaN NaN 75.0 80.0
54 2005 NaN NaN 75.0 80.0
(55, 5)

[42]: #5. MERGE MULTIPLE TABLES
#combine Students + Marks + Contact
#combining 3 datasets together
student_completespd.merge(
    students,
    marks,
    on="Student_ID",
    how="left"
)
student_completespd.merge(
    student_complete,
    contact,
    on="Student_ID",
    how="left"
)
print(student_complete.head())

Student_ID Name Department Maths Python
0 1001 Student_1 CSE 61.0 56.0
1 1002 Student_2 IT 62.0 57.0
2 1003 Student_3 ECE 63.0 58.0
3 1004 Student_4 EEE 64.0 59.0
4 1005 Student_5 MECH 65.0 60.0

[39]: #6. CONCAT ROW-WISE
#split students into two parts.
part1=students.iloc[:25]
part2=students.iloc[25:]
combined_rows =pd.concat([part1,part2])

```

JupyterLab Python [conda env:base] Anaconda Toolbox

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```

print(combined_rows.shape)
(50, 3)

[41]: #7. Concat column - wise
student_basics=students[['Student_ID','Name']]
student_marks=marks[['Python']]
combined_colspd.concat(
    [student_basics,student_marks],
    axis=1
)
print(combined_cols.head())

Student_ID Name Python
0 1001 Student_1 56
1 1002 Student_2 57
2 1003 Student_3 58
3 1004 Student_4 59
4 1005 Student_5 60

[3]: ## JOIN
#join() works using indexes
import pandas as pd
studentspd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_names="students")
marks=pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_names="marks")
contactspd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_names="contact")
students_index=students.set_index("Student_ID")
contact_index=contact.set_index("Student_ID")

joined_df=students_index.join(contact_index)
print(joined_df.tail(10))

Name Department Email
Student_ID
1041 Student_41 CSE student41@gmail.com

```

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JupyterLab Python [conda env:base] Anaconda Toolbox

```
#join() works using indexes
import pandas as pd
students=pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_names="students")
marks=pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_names="marks")
contacts=pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_names="contact")
students_index=students.set_index("Student_ID")
contacts_index=contacts.set_index("Student_ID")

joined_df=students_index.join(contacts_index)
print(joined_df.tail(10))
```

Student_ID	Name	Department	Email
1041	Student_41	CSE	student41@college.edu
1042	Student_42	IT	student42@college.edu
1043	Student_43	ECE	student43@college.edu
1044	Student_44	EEE	student44@college.edu
1045	Student_45	MECH	student45@college.edu
1046	Student_46	CSE	student46@college.edu
1047	Student_47	IT	student47@college.edu
1048	Student_48	ECE	student48@college.edu
1049	Student_49	EEE	student49@college.edu
1050	Student_50	MECH	student50@college.edu

[1]

PIVOT TABLE

```
jupyter pivot table DAY-4 Last Checkpoint: 22 hours ago
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JupyterLab Python [conda env:base] Anaconda Toolbox

[3]: # A Pivot Table is used to summarize and analyze data by grouping rows and
# Applying aggregate function such as sum,mean,count,max,min,etc
# Step-1: Load Dataset
import pandas as pd
df=pd.read_csv("people-100.csv")
print(df.head())
# to knw only the columns
print(df.columns)

      Index      User Id First Name Last Name  Sex \
0         1  88F783d2bcf9f5    Shelby  Terrell  Male
1         2  f90c03e76f1A9b9    Phillip  Summers Female
2         3  0beAb8CcdeFC2c    Kristine   Travis  Male
3         4  A31Bee3c201ef58    Yesenia  Martinez Male
4         5  1bA7A3dc874da3c        Lori     Todd  Male

      Email      Phone Date of birth \
0  elijah5@example.net  001-084-906-7849x73518  1945-10-26
1  bethany14@example.com  214.112.6044x4913  1910-03-24
2  bthompson@example.com  277.609.7938  1992-07-02
3  kaitlinkaiser@example.com  584.094.6111  2017-08-03
4  buchananmanuel@example.net  689-207-3558x7233  1938-12-01

      Job Title
0  Games developer
1  Phytotherapist
2  Homeopath
3  Market researcher
4  Veterinary surgeon
Index(['Index', 'User Id', 'First Name', 'Last Name', 'Sex', 'Email', 'Phone',
      'Date of birth', 'Job Title'],
      dtype='object')
```

```
jupyter pivot table DAY-4 Last Checkpoint: 22 hours ago
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Index(['Index', 'User Id', 'First Name', 'Last Name', 'Sex', 'Email', 'Phone',
      'Date of birth', 'Job Title'],
      dtype='object')

[4]: #Step-2 Create Age Column
df['Date of birth']=pd.to_datetime(df['Date of birth'])
today=pd.Timestamp.today()
#(timestamp.today() is used to calculate today's time
df['Age']=
    (today-df['Date of birth']).dt.days//365
)
print(df[['First Name','Sex','Age']].head())

      First Name  Sex  Age
0      Shelby  Male   80
1    Phillip  Female  116
2    Kristine   Male   33
3    Yesenia   Male    8
4        Lori   Male   87

[5]: #1.
pivot1=pd.pivot_table(
    df,
    values='Age',
    index='Sex'
)
print(pivot1)

      Age
Sex
Female  64.981132
Male    54.957447

[5]: #2.Pivot Table with aggfunction='mean'
# all the aggregate function works here
```

```
jupyter pivot table DAY-4 Last Checkpoint: 22 hours ago
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[5]: #2.Pivot Table with aggfunction='mean'
# all the aggregate function works here

pivot2=pd.pivot_table(
    df,
    values='Age',
    index='Sex',
    aggfunc='mean'
)
print(pivot2)

      Age
Sex
Female  64.981132
Male    54.957447

[6]: #3. using sum()
pivot2=pd.pivot_table(
    df,
    values='Age',
    index='Sex',
    aggfunc='sum'
)
print(pivot2)

      Age
Sex
Female  3444
Male    2583

[7]: #4. using count()
pivot2=pd.pivot_table(
    df,
    values='Age',
    index='Sex',
    aggfunc='count'
)
```


Jupyter pivot table DAY-4 Last Checkpoint: 22 hours ago

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```
[7]: #4. using count()
pivot2=pd.pivot_table(
    df,
    values='Age',
    index='Sex',
    aggfunc='count'
)
print(pivot2)
```

	Age
Sex	
Female	53
Male	47

```
[8]: #5. Multiple Aggregations
pivot5pd.pivot_table(
    df,
    values='Age',
    index='Sex',
    aggfunc=['mean','max','min']
)
print(pivot5)
```

	mean	max	min
	Age	Age	Age
Sex			
Female	64.981132	117	8
Male	54.957447	117	5

```
[10]: #6. Adds overall average
pivot6=pd.pivot_table(
    df,
    values='Age',
    index='Sex',
    aggfunc='mean',
    margins=True
)
print(pivot6)
```

	Age
Sex	
Female	64.981132
Male	54.957447
All	60.270000

```
[11]: #7 Group by Job Title and Gender
#Average age by Job Title and Gender
pivot7pd.pivot_table(
    df,
    values='Age',
    index='Job Title',
    columns='Sex',
    aggfunc='mean'
)
print(pivot7)
```

	Female	Male
Sex		
Job Title		
Accountant, chartered management	95.0	88.0
Accounting technician	17.0	NaN
Advertising account executive	18.0	NaN
Agricultural consultant	NaN	34.0
Air broker	84.0	NaN

Jupyter pivot table DAY-4 Last Checkpoint: 22 hours ago

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```
[10]: #6. Adds overall average
pivot6pd.pivot_table(
    df,
    values='Age',
    index='Sex',
    aggfunc='mean',
    margins=True
)
print(pivot6)
```

	Age
Sex	
Female	64.981132
Male	54.957447
All	60.270000

```
[11]: #7 Group by Job Title and Gender
#Average age by Job Title and Gender
pivot7pd.pivot_table(
    df,
    values='Age',
    index='Job Title',
    columns='Sex',
    aggfunc='mean'
)
print(pivot7)
```

	Female	Male
Sex		
Job Title		
Accountant, chartered management	95.0	88.0
Accounting technician	17.0	NaN
Advertising account executive	18.0	NaN
Agricultural consultant	NaN	34.0
Air broker	84.0	NaN

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```
[13]: #Count people by Job and Gender
#Above we got none so to avoid that i am giving it as fill_value
pivot8pd.pivot_table(
    df,
    values='User Id',
    index='Job Title',
    columns='Sex',
    aggfunc='count',
    fill_value=0
)
print(pivot8)
#nan fill value with 0
```

	Female	Male
Sex		
Job Title		
Accountant, chartered management	1	1
Accounting technician	1	0
Advertising account executive	1	0
Agricultural consultant	0	1
Air broker	1	0
...		
Travel agency manager	0	1
Veterinary surgeon	0	1
Warden/ranger	0	1
Warehouse manager	0	1
Waste management officer	0	1

Data Cleaning

```
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[12]: !pip install pandas
Requirement already satisfied: pandas in c:\users\danush pranav\anaconda3\lib\site-packages (2.3.3)
Requirement already satisfied: numpy>=1.26.0 in c:\users\danush pranav\anaconda3\lib\site-packages (from pandas) (2.3.5)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\danush pranav\anaconda3\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\danush pranav\anaconda3\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\danush pranav\anaconda3\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: six>=1.5 in c:\users\danush pranav\anaconda3\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)

[38]: import pandas as pd
df=pd.read_excel("Data_Cleaning_50Plus_Records.xlsx")

[58]: #2. Renaming Columns
#Check current Column Names
print(df.columns)

Index(['Student_ID', 'Student_Name', 'Department', 'Age', 'Salary', 'Email'], dtype='object')

[59]: #Rename Columns
df.rename(columns={
    'Student_ID':'Student_ID',
    'Student_Name ':'Student_Name',
    'Department ':'Department'
},inplace=True)
print(df.columns)
print(df.dtypes)

Index(['Student_ID', 'Student_Name', 'Department', 'Age', 'Salary', 'Email'], dtype='object')
Student_ID    object
Student_Name  object
Department    object
```

```
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Age      int64
Salary   float64
Email    object
dtype: object

[60]: #Convert Student_ID to String
df['Student_ID']=df['Student_ID'].astype(str)
#Convert age to Integer
#First handle missing values
df['Age']=df['Age'].fillna(0)
df['Age']=df['Age'].astype(int)
#check
print(df.dtypes)

Student_ID    object
Student_Name  object
Department    object
Age           int64
Salary        float64
Email         object
dtype: object

[61]: #4.Detecting Duplicates records
#find duplicates
duplicates=df.duplicated()
print(duplicates)

0    False
1    False
2    False
3    False
4    False
5    False
6    False
7    False
```

```
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19    False
20    False
21    False
22    False
23    False
24    False
25    False
26    False
27    False
28    False
29    False
30    False
31    False
32    False
33    False
34    False
35    False
36    False
37    False
38    False
39    False
40    False
41    False
42    False
43    False
44    False
45    False
46    False
47    False
48    False
49    False
50    False
51    False
dtype: bool
```